Liquidity Event Prediction Model

Track D — listed promoter & insider sell-down · build and backtest results

Built on live NSE exchange data. No simulated data anywhere in this build. Run date 7 August 2026.

Headline	
Cash-dated events in register	4,749
Panel	100,630 entity-months · 2,054 companies
Precision@50 (6m horizon)	0.302 vs 0.109 base rate
Lift over base rate	2.78×
Median lead time	6.0 months
Best baseline beaten	Largest-100 (p@50 0.228)

What this covers — and what it does not

The framework specifies six event tracks. **This build covers Track D end to end on real data, plus the deterministic part of Track A (post-listing lock-in calendar).** Tracks A stage 1–2, B, C, E and F are specified in code but **not backtested**, because they need MCA21, private-market and rating-agency data that is not reachable without a commercial contract.

This matters for how the numbers should be read: the framework itself calls Track D "the most crowded and lowest-lead-time" track, and locates the real edge in Track A (pre-DRHP) and Track C. **The track that has been backtested is the one the framework rates as least differentiated.** That is a data-access outcome, not a judgement about where the value is.

1. What is built

Layer	Status	What it does
L0 Entity resolution	Built	50,347 promoter/insider identities across 1,957 symbols
L1 Ingestion	Built	Cached, rate-limited NSE client; 7 sources
L2 Point-in-time store	Built	Bitemporal: valid_from vs known_from, revision-aware
L3 Event register	Built	4,749 cash-dated events, 3 independent sources
L3 Features	Partial	3 of 8 framework families available from exchange data
L4 Hazard model	Built	Discrete-time competing risks, multinomial logit
L5 Wallet sizing	Built	FY26-27 tax config; deterministic for listed events
L6 Ranking	Built	Probability vs expected-proceeds objectives
T3 Rules engine	Built & live	389 current alerts, each citing its regulation
Backtest harness	Built	Walk-forward, 12m embargo, per-month ranking

Data ingested

Source	Rows	Span
Bulk + block deals	157,029	2015-01 → 2026-08
Shareholding patterns	36,408	2015-12 → 2026-07
PIT Reg 7(2) insider disclosures	24,146	2008-10 → 2026-08
SAST Reg 29	40,980	2018 → 2026
IPO history (lock-in calendar)	1,346	—
Trading-plan disclosures	136	2021 → 2026

A trap worth recording: the NSE bulk/block-deal JSON endpoint silently truncates every query to 70 rows regardless of date range. Trusting it would have produced ~9,800 deals instead of 157,029 — a 94% loss of the label set, with no error raised. Only `&csv=true` returns the full set.

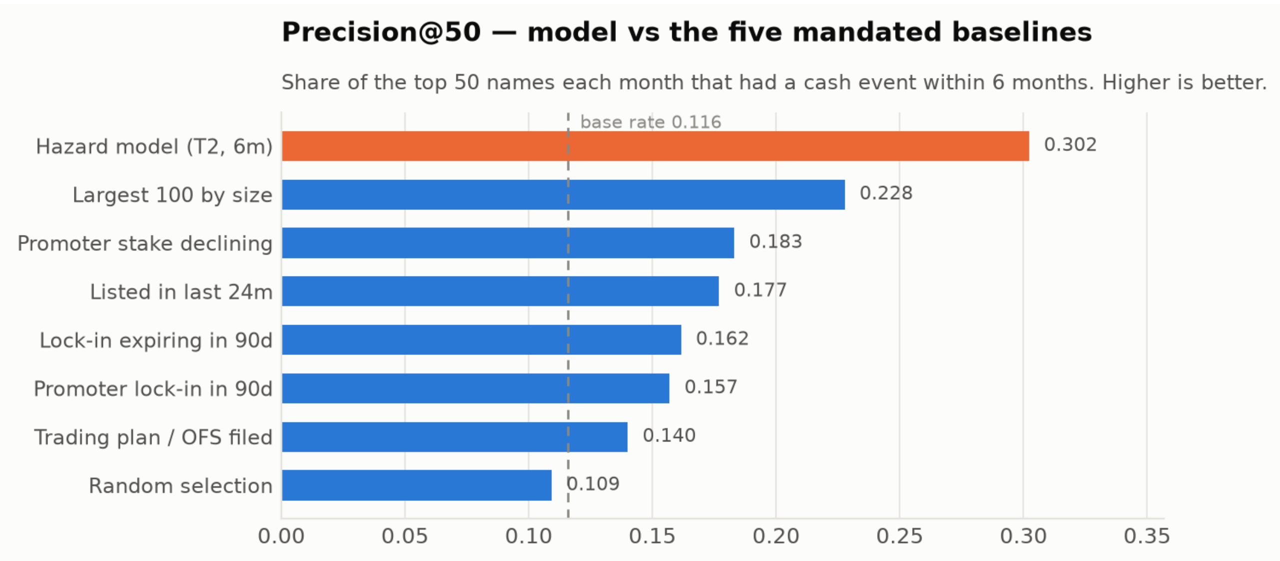
2. Model results

Walk-forward, expanding window, with two separate leakage controls: training stops at *test start – horizon* so training labels cannot resolve inside the test period, and then a further 12-month embargo on top. Ranking is done **within each test month**, because precision@50 means "the 50 names an RM team can work this month".

Tier	Horizon	Folds	p@25	p@50	p@100	Lift@50	AUC	Cal. err
T1	3m	4	0.232	0.216	0.172	3.48	0.659	0.038
T2	3m	4	0.245	0.208	0.182	3.29	0.654	0.038
T3	3m	4	0.207	0.199	0.187	3.12	0.656	0.037
T1	6m	3	0.342	0.312	0.251	2.83	0.646	0.063
T2	6m	3	0.347	0.302	0.251	2.78	0.645	0.062
T3	6m	3	0.342	0.294	0.245	2.68	0.642	0.060
T1	12m	1	0.520	0.460	0.400	2.51	0.647	0.146
T2	12m	1	0.480	0.420	0.390	2.29	0.641	0.153
T3	12m	1	0.520	0.440	0.380	2.40	0.640	0.153

More features made it monotonically worse. T1 (15 features) ≥ T2 (24) ≥ T3 (30) at every horizon — the late confirmatory signals reserved for T3 add nothing. The tier structure earns its keep as a governance device (T1 provably excludes late signals), not as a performance ladder.

Against the five mandated baselines



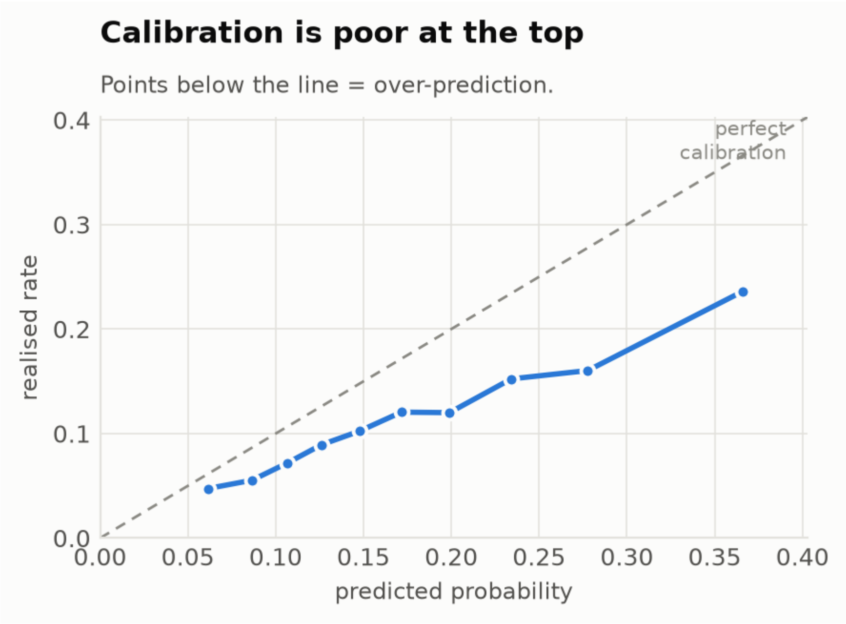
The model beats every baseline on hit rate. It loses decisively to "just call the 100 biggest companies" on rupees captured — see section 3.

Lead time — the metric the business actually cares about

Score threshold	Events flagged in advance	Median lead	IQR
top decile	468	6.0 months	3.4 – 11.0
top 5%	325	5.9 months	3.0 – 9.3
top 1%	137	6.5 months	4.0 – 10.8

A median ~6 months of warning is commercially real — it lands inside the framework's T2 "active coverage, tax and estate conversations" window.

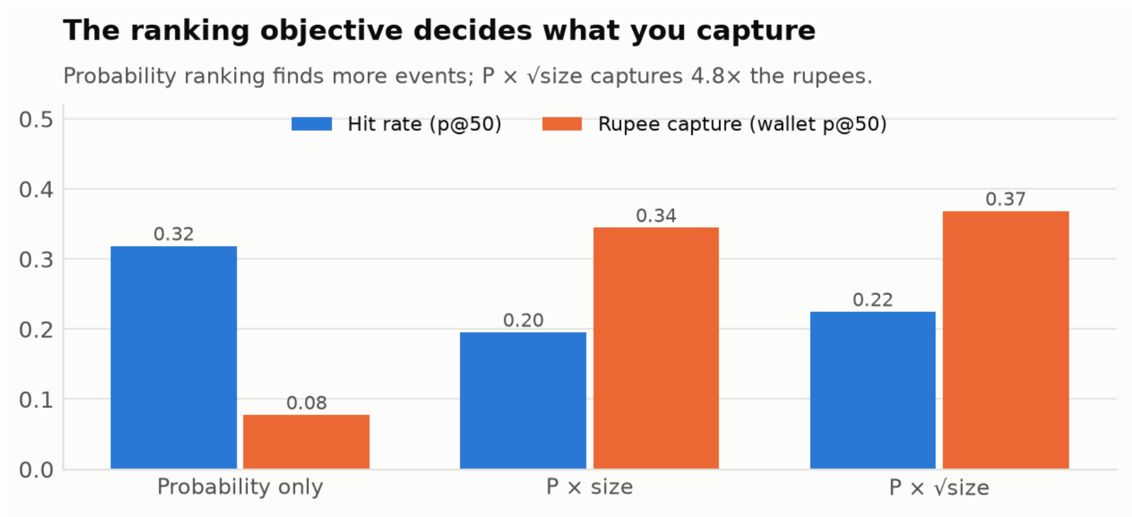
Where it is weak



The model over-predicts by roughly 70% in the top decile (predicted 0.369 vs realised 0.217). The framework is explicit that "a well-calibrated 30% must mean 30%, or wallet-weighted prioritisation is garbage". Cause is regime shift between train and test periods. **This needs isotonic recalibration on a held-out fold before any production use.**

3. The ranking objective was wrong by default

The single most commercially consequential finding. Ranking by probability alone loses to a trivial size baseline on rupees captured. Re-ranking on expected proceeds — P(event) × E[size | event], estimated point-in-time — fixes it.



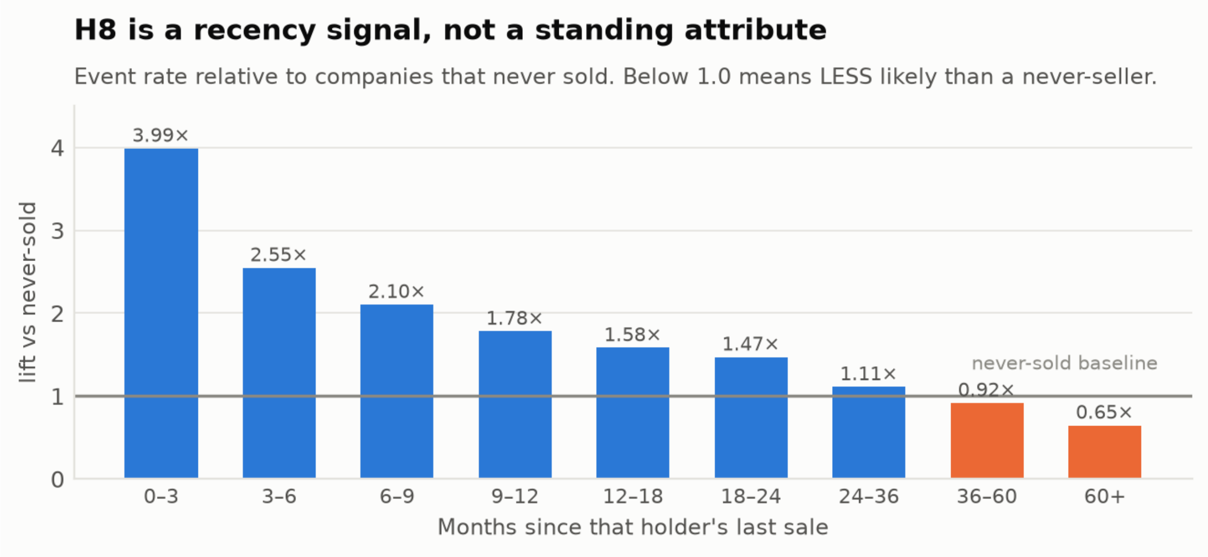
The damped version (P × √size) captures **4.8× the rupees** of probability ranking for a ~30% reduction in hit rate, and beats the largest-100 baseline (0.307) as well. Since RM capacity is the binding constraint and one Rs 500 cr event is worth twenty Rs 10 cr ones, **this is the objective the list should ship on.**

4. Hypothesis results

#	Hypothesis	Prior	Verdict	Measured
H1	Lock-in expiry raises P(sell in 90d)	High	NOT SUPPORTED	1.51× raw, but 0.86× once controlled for the recent-IPO cohort (p=0.0099)
H2	Trading plan followed by actual selling	High	NOT AT THAT STRENGTH	25% follow-through [12–45%], not "the great majority"
H8	Prior liquidity predicts repeat events	Medium	SUPPORTED	1.49× lift, p=2.8e-195, monotonic in prior count
H3–H7, H9, H10	Track A / B / C hypotheses	Mixed	NOT TESTABLE	Require MCA21, private-market and rating-agency data

H1 is the cautionary result. The uncontrolled test returned 1.51× at p=4.7e-08 — significant, right sign, matching its "High" prior. But lock-in expiries cluster 6 and 18 months after listing, so "lock-in just expired" is largely a proxy for "recently listed". Comparing recent IPOs with a just-expired lock-in against other recent IPOs, the lift falls **below 1**. The mechanism is real and the calendar is deterministic, but as a standalone ranking signal it adds nothing over "this company listed recently".

H8 refined: it is recency, not type



A holder who sold five years ago is **less** likely to sell than one who never sold. So the "ever sold" flag is the wrong feature — this is a recency signal with roughly a three-year half-life. Inter-event gaps are also over-dispersed (median 101 days, CV 1.56), meaning selling is **bursty, not regular**: the Rule-144 dribble-out analogy the framework imports from the US does not transfer, because India imposes no equivalent volume cap. There are no predictable tranches to calendar.

5. Model result by company — backtest

Every company the model scored in a held-out test month. "Surfaced" means the model ranked it in the top 50 that month; "precision when surfaced" is how often a cash event actually followed within 6 months when it did. Sorted by realised wallet captured.

Symbol	Company	Times surfaced	Hits	Precision when surfaced	Wallet captured (Rs cr)	Max score
VMM	Vishal Mega Mart Limited	8	6	75%	30,588	0.59
PREMIERENE	Premier Energies Limited	8	5	62%	13,145	0.58
SAGILITY	SAGILITY LIMITED	5	4	80%	10,981	0.57
INDIGO	InterGlobe Aviation Limited	3	3	100%	4,396	0.46
PRIVISCL	Privi Speciality Chemicals Limited	3	2	67%	4,119	0.44
ONESOURCE	Onesource Specialty Pharma Limited	6	5	83%	3,645	0.52
AETHER	Aether Industries Limited	5	5	100%	3,281	0.46
NETWEB	Netweb Technologies India Limited	9	5	56%	3,181	0.53
YATHARTH	Yatharth Hospital & Trauma Care Se	11	7	64%	2,424	0.48
ANANDRATHI	Anand Rathi Wealth Limited	3	3	100%	2,120	0.41
KAYNES	Kaynes Technology India Limited	10	5	50%	1,735	0.49
BLACKBUCK	BLACKBUCK LIMITED	7	6	86%	1,436	0.57
MOBIKWIK	One Mobikwik Systems Limited	13	9	69%	1,133	0.60
AWFIS	Awfis Space Solutions Limited	10	8	80%	1,067	0.58
INDGN	Indegene Limited	5	2	40%	903	0.50
ATHERENERG	Ather Energy Limited	3	1	33%	631	0.46
OLAELEC	Ola Electric Mobility Limited	7	7	100%	551	0.61
AVALON	Avalon Technologies Limited	10	4	40%	549	0.51
SIGNPOST	Signpost India Limited	6	6	100%	450	0.52
POCL	Pondy Oxides & Chemicals Limited	13	10	77%	342	0.49
PARAS	Paras Defence and Space Technologi	3	3	100%	317	0.48
AGIIL	Agi Infra Limited	1	1	100%	241	0.42
SHAILY	Shaily Engineering Plastics Limite	1	1	100%	170	0.38
AARTIPHARM	Aarti Pharmalabs Limited	8	5	62%	123	0.48
NAZARA	Nazara Technologies Limited	6	1	17%	112	0.42
MASTERTR	Master Trust Limited	7	7	100%	95	0.50
AAATECH	AAA Technologies Limited	1	1	100%	52	0.38
MOSCHIP	Moschip Technologies Limited	5	5	100%	29	0.53
SUYOG	Suyog Telematics Limited	4	1	25%	26	0.53
AFIL	Akme Fintrade (India) Limited	7	4	57%	16	0.52

Full table for all 1,927 scored companies: `reports/company_backtest_results.csv`. Month-by-month scores for every company: `reports/company_scores_backtest.csv`.

6. Live origination ranking

Model refit on all months whose labels have fully resolved, scored forward as of July 2026. Ranked on P × √size — the objective from section 3. This is the list an RM team would work.

These are model outputs on public data, not recommendations, and not a claim that any named party intends to sell. Handle as sensitive internal research.

#	Symbol	Company	P(event 6m)	Exp. gross (Rs cr)	Exp. addressable AUM (Rs cr)	Prom. %	Mths since last sale
1	AADHARHFC	Aadhar Housing Finance Limited	0.094	13,863	453	64.9	3.0
2	VMM	Vishal Mega Mart Limited	0.137	5,102	242	40.1	3.0
3	BHARTIARTL	Bharti Airtel Limited	0.099	7,262	250	50.1	6.0
4	ANTHEM	Anthem Biosciences Limited	0.216	1,275	95	74.7	0.4
5	ENRIN	Siemens Energy India Limited	0.134	2,928	136	75.0	6.5
6	SAGILITY	SAGILITY LIMITED	0.111	3,660	140	51.0	6.0
7	JSWSTEEL	JSW Steel Limited	0.116	3,150	126	45.3	1.4
8	BAJAJHFL	Bajaj Housing Finance Limited	0.130	2,294	104	86.7	6.0
9	LODHA	Lodha Developers Limited	0.168	1,100	64	72.3	0.2
10	CORONA	CORONA Remedies Limited	0.193	749	50	69.0	0.5
11	ADANIGREEN	Adani Green Energy Limited	0.087	3,519	106	62.4	36.0
12	COHANCE	Cohance Lifesciences Limited	0.088	3,094	95	57.5	9.0
13	ADANIENT	Adani Enterprises Limited	0.075	4,140	107	74.7	18.0
14	VEDL	Vedanta Limited	0.084	2,940	85	56.4	0.3
15	MPHASIS	MphasiS Limited	0.062	4,726	101	30.6	6.0
16	EMCURE	Emcure Pharmaceuticals Limited	0.200	426	29	77.9	0.2
17	OLAELEC	Ola Electric Mobility Limited	0.221	322	25	33.0	0.9
18	PATANJALI	Patanjali Foods Limited	0.092	1,816	58	68.2	6.0
19	MAZDOCK	Mazagon Dock Shipbuilders Limi	0.064	3,678	82	81.2	12.0
20	INDIGO	InterGlobe Aviation Limited	0.114	1,156	46	41.6	9.0
21	SANOFICONR	Sanofi Consumer Healthcare Ind	0.065	3,465	78	71.3	20.2
22	IRFC	Indian Railway Finance Corpora	0.078	2,333	63	84.7	3.0
23	PREMIERENE	Premier Energies Limited	0.203	333	23	63.9	1.2
24	AIIL	Authum Investment & Infrastruc	0.104	1,236	45	68.8	9.0
25	JSWCEMENT	JSW Cement Limited	0.205	312	22	72.0	1.3
26	BELRISE	Belrise Industries Limited	0.118	897	37	66.5	6.0
27	TCS	Tata Consultancy Services Limi	0.053	4,495	82	71.8	27.0
28	MEESHO	Meesho Limited	0.159	494	27	16.6	0.7
29	SUMICHEM	Sumitomo Chemical India Limite	0.023	23,987	189	75.0	69.0
30	DOMS	DOMS Industries Limited	0.167	420	24	70.4	0.5

Full ranked list of all 2,054 companies: `reports/company_live_ranking.csv`. Deterministic T3 alerts (389 as of 7 Aug 2026, each citing the regulation that creates its lead time): `reports/t3_alerts.csv`.

7. What to fix before this is used

1. **Recalibrate.** Isotonic regression on a held-out fold. The top decile over-predicts by 70%, which breaks wallet-weighted prioritisation.

2. **Ship P × √size ranking**, not probability ranking — 4.8× the rupee capture.

3. **Licensed insider-trade feed.** The NSE Reg 7(2) endpoint retains only ~20 disclosures per symbol, which is the binding constraint on settling H2.

4. **Parse offer documents** for the true lock-in term before writing off H1.

5. **Then buy MCA21 access** and build Track A — the structural-readiness signals remain completely untested, and that is where the framework argues the edge lives.

Known limitations of these results

• **Panel window is 2021-07 to 2026-07** (61 months), bounded by when promoter identity becomes knowable from exchange filings — not the framework's 2015–2025 vintage split.

• **The model is mostly one feature.** Standardised coefficients are dominated by months-since-last-event and ever-sold; everything else is an order of magnitude smaller.

• **Base rates are higher than the framework assumes** (11.2% at 6m vs an assumed 2–5% per entity-year) because the register counts every disclosed insider sale above Rs 1 cr, including routine director sell-downs.

• **Lock-in dates are assumed** from the standard ICDR schedule, not read from offer documents; the 18m vs 36m MPC distinction is unresolved.

• **Naive vs point-in-time joins scored the same here** (0.307 vs 0.302). That is an honest null, not proof that PIT discipline is optional: 11.3% of entity-months get a different value under the naive join, and the shareholding feed contains 4,631 restatements with a maximum publication lag of 2,941 days. The time machine is loaded; this particular model just does not ride it.

• **propensity_to_externalise is assumed at 40%, not fitted** — it must be calibrated from the firm's own conversion history.

• **Regulatory and tax parameters were transcribed from the framework document and have not been independently re-verified** against primary sources.

Compliance

Every input is an exchange disclosure or regulatory filing published by NSE — public sources only, no non-public information. Scoped to client origination, not trading. The event register and promoter registry contain personal data on 50,347 identifiable individuals, so DPDP Act 2023 lawful basis, purpose limitation and retention need to be established before go-live.